

Hadoop Eco-System (Hive)



Agenda

- Hive Introduction
- Hive Features
- Hive Applications
- Practical's

HIVE

- Hive is a data warehouse infrastructure tool to process structured data in Hadoop
- It resides on top of Hadoop to summarize Big Data, and makes querying and analyzing easy.
- Hive is not
 - A relational database
 - A design for Online Transaction Processing (OLTP)
 - A language for real-time queries and row-level updates

HQL (Hive Query Language)

How is it Different from SQL

- The major difference is that a Hive query executes on a Hadoop infrastructure rather than a traditional database.
- This allows Hive to scale to handle huge data sets - data sets so large that high-end, expensive, traditional databases would fail.
- The internal execution of a Hive query is via a series of automatically generated Map Reduce jobs

Hive Features

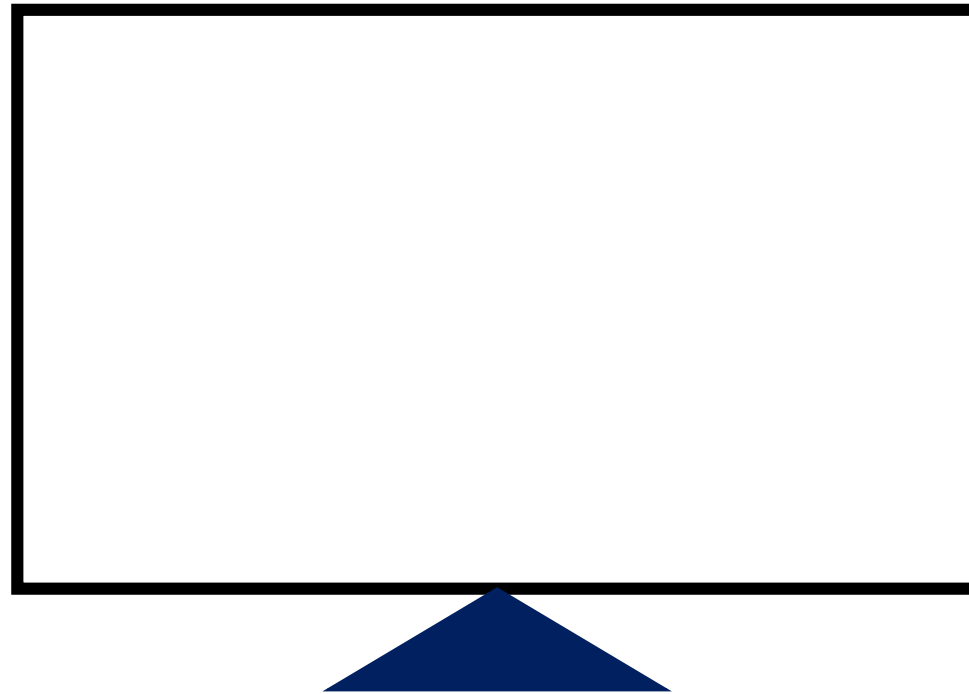
- Hive query Language is very similar to SQL.
- Hive does not have SQL statements such as:
 - INSERT
 - UPDATE
 - DELETE
- The underlying reason why these fundamental SQL statements are not available in Hive is that Hive (and Hadoop) technologies are designed for Data Warehousing applications (that is, write once, read many times). Used for OLAP

Hive Applications

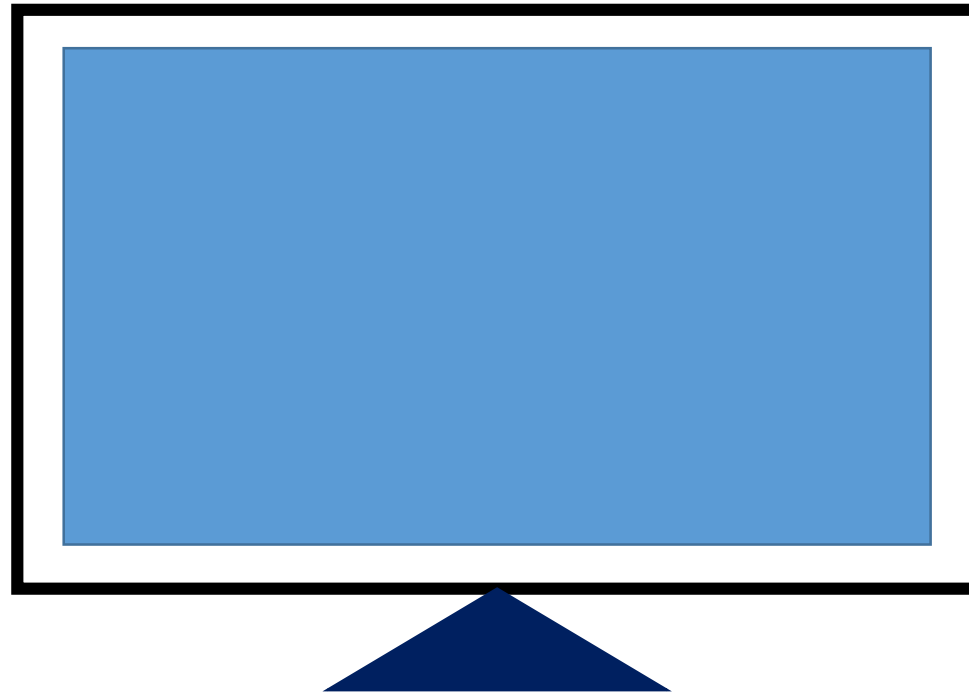
Hive Usage Scenario

- Text mining
- Log Processing
- Document indexing
- Customer-facing business intelligence (e.g., Web Analytics)

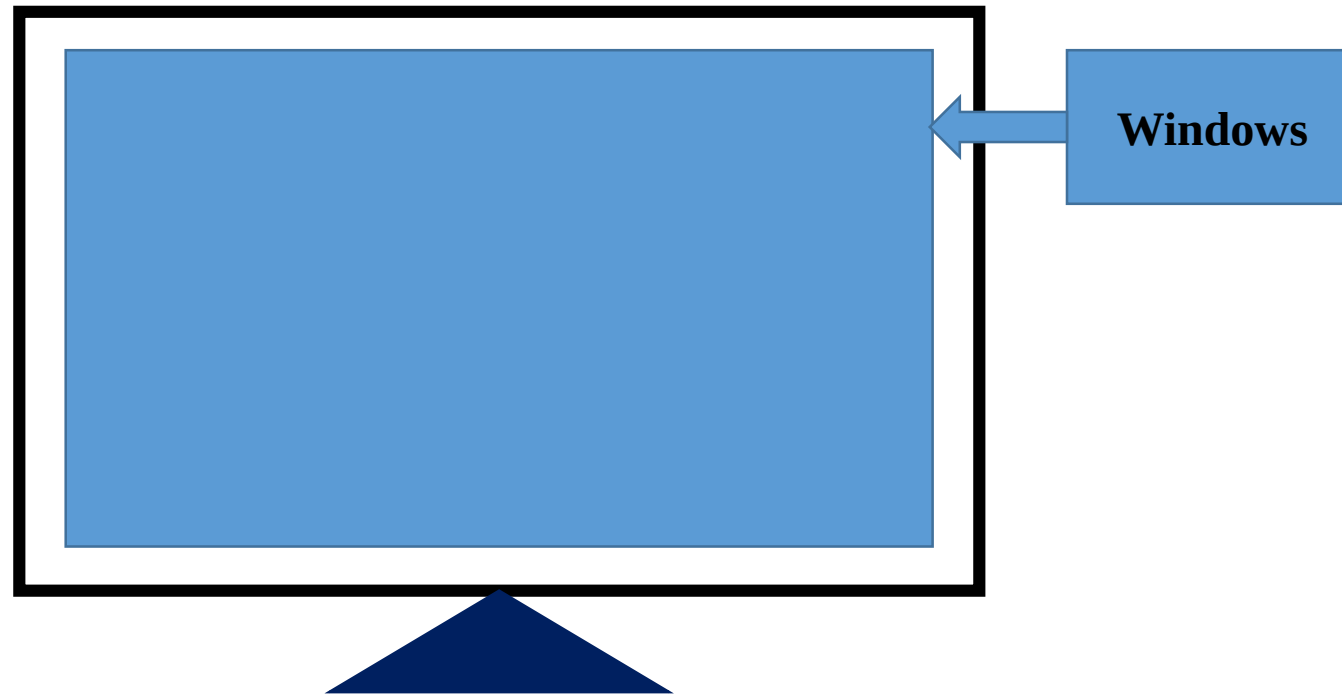
Structure (where to run jobs)



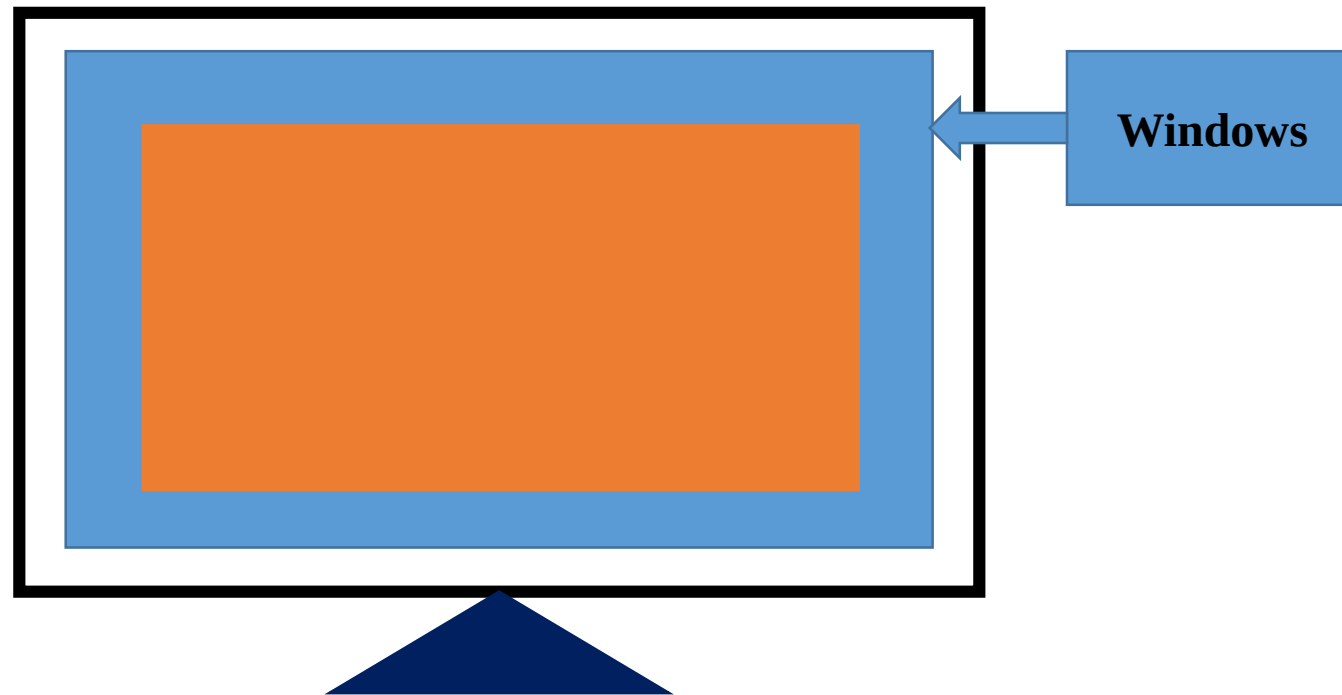
Structure (where to run jobs)



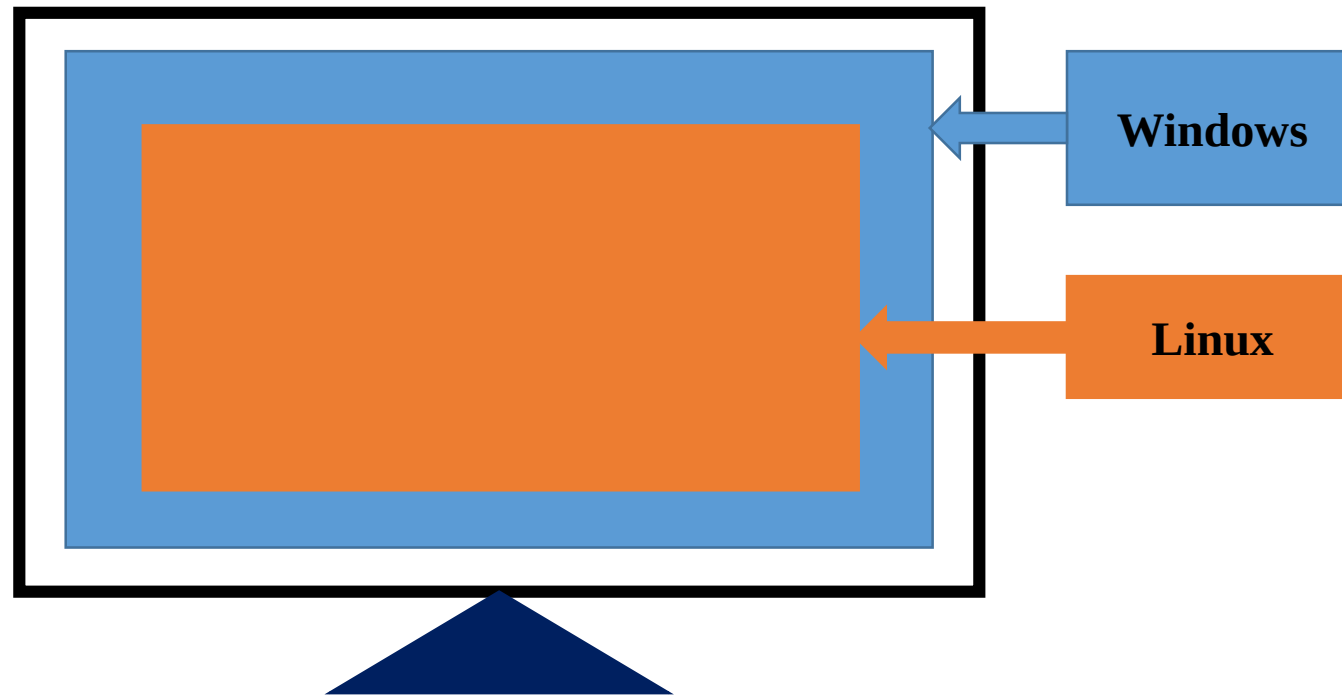
Structure (where to run jobs)



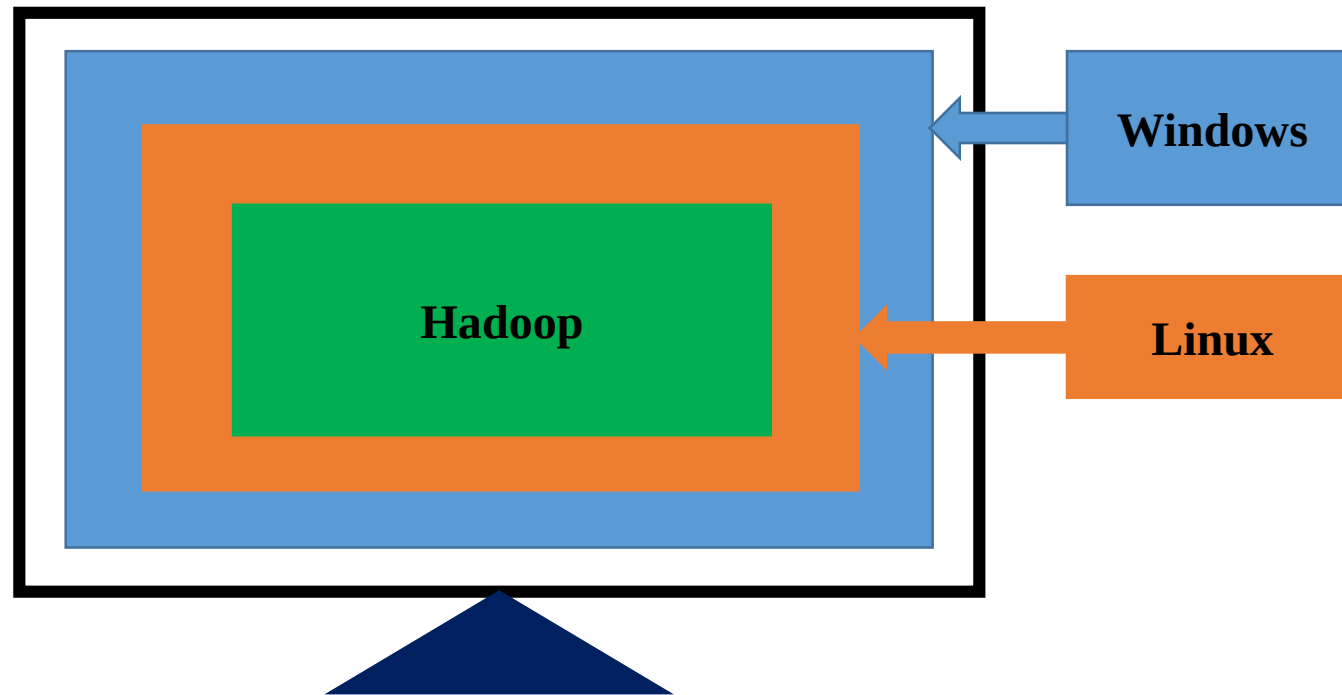
Structure (where to run jobs)



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Structure (where to run jobs)



Cluster Mode Types

- Pseudo Mode (Single Node Cluster)
 - All services running in 1 machine
 - Research and Development
 - Training /Practice
 - POC
- Cluster Mode (Multi Node Cluster – Fully Distributed System)
 - Real time Projects
 - Clients
 - DEV,UAT & PROD

Linux Commands

- cal
- date
- ls
- who
- mkdir
- cp
- vi

- Etc.... (browse for Linux commands) {not required for us}

Hadoop Commands

- Hadoop fs -ls
- Hadoop fs -mkdir
- Hadoop fs -cat
- Hadoop fs -copyFromLocal
- Hadoop fs -moveFromLocal
- Hadoop fs -cp
- Hadoop fs -mv

Thank You !!!

Questions??